

WEB APPLICATIONS AND SECURITY

REVISION NOTES

Computer Accessibility refers to the user friendliness of a computer system for all, regardless of their disability. It enables a person with a disability or impairment to use a computer. It is known as Assistive Technology. There are numerous types of impairment that impact computer usage. These include:

- Cognitive impairments and learning disabilities
- Attention Deficit Hyperactivity Disorder (ADHD) or autism.
- Visual impairment.
- Hearing impairment
- Motor impairment

Launching Accessibility Options:

To launch accessibility options in WindowsXP, Click Start > Control Panel > Accessibility Options.

Various tabs in Accessibility Option Window:

Keyboard Tab to configure accessibility options for Keyboard is displayed

1) Sticky Keys: Sticky Keys is an accessibility feature to help computer users with physical disabilities, but it is also used by others as a means to reduce repetitive strain.

StickyKeys allows the user to press and release a modifier key, such as Shift, Ctrl, Alt, or the Windows key, and have it remain active until any other key is pressed.

- To enable StickyKeys, select Use StickyKeys.
- Click Apply.
- Click OK.

The Sticky Keys icon appears in the system tray.

2) Filter Keys: Filter Keys is a feature of Microsoft Windows. It is an accessibility function that tells the keyboard to ignore brief or repeated keystrokes, making typing easier for people with hand tremors.

- To enable Filter Keys, check Use Filter Keys.
- Click Settings under Filter Keys and check "Ignore Repeated Keystrokes".
- Click Apply.
- Click OK.

The FilterKeys icon appears in the system tray

3) ToggleKeys: ToggleKeys is an accessibility function which is designed for people who have vision impairment or cognitive disabilities. When ToggleKeys is turned on, computer emits sound cues when the locking keys (Caps Lock, Num Lock, or Scroll Lock) are pressed.

A high sound is emitted when the keys are switched on and a low sound is emitted when they are switched off.

- To enable ToggleKeys, Check Use ToggleKeys.
- Click Settings under ToggleKeys. • Click Apply
- Click OK.

Having enabled ToggleKeys, you can see its effect by any one of the following keys NUM LOCK, CAPS LOCK or SCROLL LOCK. Notice that on doing so, you hear a beep.

Sound Tab to configure accessibility options for sound is displayed.

1) SoundSentry: SoundSentry is designed to help users with auditory impairments. SoundSentry generates visual warnings, such as a blinking title bar or a flashing border, whenever the computer generates a sound.

- To enable SoundSentry, Check Use SoundSentry under Sound Tab. Doing so activates a drop down list wherefrom you can choose the type of visual warning desired.
- Click Apply.
- Click OK.

2) ShowSounds: ShowSounds instructs applications that convey information by sound, to also provide information visually, through text captions or informative icons.

- To enable ShowSounds, Check Use ShowSounds under Sound Tab.
- Click Apply.
- Click OK

Display Tab to configure accessibility options for Display is displayed.

1) High Contrast: High Contrast is an accessibility feature to assist people with vision impairment. You can change the size and color of fonts and the background for ease of viewing.

- To enable HighContrast, Check Use HighContrast
- Under Display Tab (figure 5).
- Click Apply.
- Click OK.

2) Cursor Options: Cursor Options is also an accessibility feature that assists people with vision impairment by changing the blink rate and width of the cursor.

- To change the speed of the cursor blink, move the Blink Rate slider back and forth. Notice the cursor blinking at various speeds.
- To change the width of the cursor, move the Width slider back and forth. Notice the with of the cursor changing.

Mouse Tab to configure accessibility options for Mouse is displayed.

1) MouseKeys: MouseKeys is an accessibility feature that assists people who have difficulty using a mouse. This option uses the keyboard (especially numeric keypad) as a pointing device instead of a mouse.

- To enable MouseKeys, Check Use MouseKeys.
- Click Apply.
- Click OK.

General Tab enables you to configure accessibility options for all users.

1) Serial Keys: Serial Keys is an accessibility feature that assists people that have difficulty using a keyboard or a mouse (or both). They can use special devices such as Sip, Puff and Breath Switches to provide input to the computer through Serial Ports.

NETWORKING FUNDAMENTALS

A network can be defined as an interconnected collection of autonomous computers. A ‘computer network’ or simply a ‘network’ is a collection of computers and other hardware devices, interconnected by communication channels (satellites or cables) that allow sharing of resources and information. A computer networking is the practice for exchanging informations/services between two or more computer devices together for the purpose of sharing data. The speed of a network is measured in Mbps (Megabits per second).

Networks are designed using the following architecture:

1) Peer-to-Peer (P2P) Architecture: Networks in which all computers have an equal status are called peer to peer networks. Generally in such a network each terminal has an equally competent CPU.

2) Client-Server Architecture: Networks in which certain computers have special dedicated tasks, providing services to other computers (in the network) are called client server networks. The

computer(s) which provide services are called servers and the ones that use these services are called clients.

BENEFITS OF NETWORKING

Computer network is very useful in modern environment, so some of the benefits of networking are discussed here:

- **Data Sharing:** One of the most important uses of networking is to allow the sharing of data.
- **File Sharing:** Networking of computers helps the users to share data files.
- **Hardware Sharing:** Users can share devices such as printers, scanners, CD-ROM drives, hard drives, etc.
- **Application Sharing:** Applications can be shared over the network and this allows to implement client/server applications.
- **User Communication:** This allows users to communicate using E-mail, newsgroups, video conferencing within the network.
- **Access to Remote Database:** By networking, we are able to access to the remote database. It is easy for any person using his PC to make reservations for aeroplanes, trains, hotels, etc., anywhere in the world with instant confirmation within the network.

TYPES OF NETWORK

A network refers to a group of interconnected computers which are capable of sharing information and communication devices.

On the basis of coverage or geographical spread, a network can be divided into following types:

LAN (Local Area Network)

When a group of computers and its devices are connected in a small area, then this network is said to be a LAN. Computers or users in a local area network can share data, information, software and common hardware devices such as printer, modem, hard disk, etc. A LAN typically relies mostly on wired connections for increased speed and security but wireless connection also be a part of LAN. LAN are used within office building, school, etc.

MAN (Metropolitan Area Network)

This is basically bigger version of LAN and normally uses similar technology. It might cover few buildings in a city and might either be private or public. e.g. In a city, a MAN, which can support both data and voice might even be related to local cable television network.

WAN (Wide Area Network)

The network which connects the different countries network is known as WAN. It can be a group of LANs. The largest existing WAN is Internet. For example, a network of ATMs, BANKS, National Government Offices spread over a country or continents are examples of WANs.

INTERNET:

The Internet is a global system of interconnected computer networks that use the standard Internet protocol suite to serve billions of users worldwide. It is a network of networks.

Uses of Internet: There are various uses of Internet. Some of them are listed below

- Internet is used by students, educational institutes to gather information for research.
- It is used for online shopping.
- It is used for sending and receiving mail.
- It is used for playing games.
- It is used for Online Transaction.

World Wide Web: World Wide Web (abbreviated as WWW or W3, commonly known as the Web), is a system of interlinked hypertext documents accessed via the Internet.

Web Browser: A Web Browser is a software used to view Web sites and acts as an interface between the user and the World Wide Web.

Web Server: Web server is a computer that stores web sites and their related files for viewing on the Internet.

Internet Service Provider: An Internet service provider (ISP) is an organization which provides you with access to the Internet via a dial-up (using modem) or direct (hard wired) or wireless connection. for example Bharat Sanchar Nigam Limited (BSNL), Airtel, MTS, Vodafone, Tata Docomo, etc.

Modem: It stands for MOdulator/DEModulator. It is a device that converts digital computer signals into a form analog signals and vice versa.

Types of Common Internet Connectivity: There are different types of Internet Connectivity available today; it can be widely categorized into wired and wireless access.

Technology	Type of Connectivity
Dial-up	Wired
DSL	Wired
Cable Internet Access	Wired
3G	Wireless
WiMAX	Wireless
Wi-Fi	Wireless

Some of the commonly used Internet connectivity are:

1) Dial-up connections are extremely slow and in most cases, it is replaced by a high speed connection such as DSL or Cable Modem.

2) DSL: Digital subscriber line(DSL) provide Internet access by transmitting digital data over wires of a local telephone network. DSL service is delivered along with wired telephone service on the same telephone line. On the customer premises, a DSL filter removes the high frequency interference, enabling simultaneous use of the telephone and data transmission. For using a DSL connection, you need a DSL modem and a subscription.

3) Cable Internet Access: Cable Internet Access is a form of broadband Internet access that uses the cable television infrastructure. Cable Internet Access is provided through existing cable TV networks; this is similar to DSL that is provided over existing telephone lines

4) 3G: 3G, short for 3rd Generation is a set of standards used for mobile devices and mobile telecommunication services and networks. High-Speed Downlink Packet Access (HSDPA) is 3G mobile telephony communications protocol that allows higher data transfer speeds and capacity.

5) WiMAX: WiMAX (Worldwide Interoperability for Microwave Access) is a wireless communications standard designed to provide mobile broadband connectivity across cities and countries through a variety of devices. WiMAX is a long range system, covering many kilometres

and is typically used where DSL or Cable Internet Access cannot be used; this could be difficult in laying out cables for home or offices located in remote locations but need access to the Internet.

6) WI-FI : Wi-Fi (Wireless Fidelity) is a popular technology that allows an electronic device such as computers or mobile phones to exchange data wirelessly over a network, including high-speed Internet connections. WiFi devices such as personal computer, smartphones, video game console, etc. can connect to a network resource such as Internet through a device called the Wireless Access Point (WAP).

Wi-Fi is used where cables cannot be run (such as old buildings, outdoor areas) to provide network and Internet access. Wi-Fi can also be used where users require mobile connectivity.

Wi-Fi connectivity is used in home & offices, hotels, college & school campus typically for Internet Access. Shopping malls, coffee shops, resorts mostly offer free or paid Wi-Fi access to the Internet for their customers.

INSTANT MESSAGING

Instant messaging (IM) is a form of communication over the Internet that offers an instantaneous transmission of text-based messages from sender to receiver.

Types of Instant Messaging Software: There are two kinds of instant messaging software – application based and Web based.

A) Application based: Application based instant messaging software is downloaded and installed on user's computer. Some of the popular instant messaging software are:

- Google Talk
- Yahoo! Messenger
- Skype
- Windows Live Messenger
- Rediff Bol, etc.

B) Web based: Web based instant messaging software is accessed using browsers such as Internet Explorer, Mozilla Firefox, Google Chrome, etc. Some of the popular web based instant messaging software are:

- Meebo
- Yahoo! Messenger for the Web
- MSN Web Messenger
- IMO, etc.

Key Features of an instant messaging are as follows:

- Text Messages can be sent to one or more person (Similar to SMS)
- Audio calling and conferencing.
- Video calling and conferencing.
- File transfers (Not limited to documents, spread sheets, audio files, video files, etc.)
- Message history (Save messages for future reference).

NOTE: To use instant messaging software, a user must have a valid instant messaging account.

GOOGLE TALK

Google Talk : Google Talk is an instant messaging service that provides both text and voice communication developed by Google Inc. Google Talk can also be used for making video calls. Google Talk is free and is available as application based as well as web based. A Gmail account is required to use Google Talk.

Steps to Chat with a contact that is already added to your contact list.

- a) Whenever your friend in the contact list is online you can see the person along with a green dot.
- b) You can start sending text chat message instantly by double-clicking on a contact the other person will see the text message and respond to your message.

General rules and etiquettes to be followed while chatting :

- 1) Messages should be short and to the point.
- 2) Always introduce yourself by name if your screen name doesn't reflect it.
- 3) Always ask if the other person has time to chat first.
- 4) Don't Type your messages in uppercase as it seems to be extremely rude.
- 5) Give people time to respond.

CREATING AND PUBLISHING WEB PAGES – BLOG

Blog: A blog is a discussion style site used by non-technical (or technical users) users for creating personal web pages. Blogs are similar to an online personal diary and simple to use.

Some of the popular websites that offer blog service for free are:

- www.wordpress.com
- www.blogger.com
- www.blog.com
- www.weebly.com
- www.blogsome.com

WordPress: WordPress is free web service that you can use to create a beautiful website or blog. WordPress has support for “themes” which can make the blog or the webpage look attractive.

How to create account in WordPress?

- 1) Open the Web Browser and type <https://signup.WordPress.com/signup/>.
- 2) Enter a unique address to your WordPress Blog. This is the address which others will use to view your blog.
- 3) Enter username and password to manage your blog.
- 4) Enter your Email Address and click on create blog.
- 5) An email will be sent (above mentioned email) for activating your blog account. Open your email and click on the activation link.
- 6) Now the blog is ready for use.

How to create post in WordPress?

- 1) Login to your WordPress Account.
- 2) To create a post, click New Post.
- 3) Type the title for your post.
- 4) Type the content which you want others to read. You can also add photos, videos, etc. to the blog using the options available in WordPress
- 5) Once you have finished typing the content, Click Publish Post to publish your content. This process is called posting content.

USING OFFLINE BLOG EDITORS

Offline Blog Editors:

Those Editors in which we can create blog when we do not have an active internet connection and publish the blog whenever internet connectivity is available are called Offline Blog Editors.

There are several free offline blog editors available such as :

- Qumana
- Windows Live Writer
- Blogdesk

- BlogJet
- MarsEdit

Online transactions: Online transactions deals with transfer of money over the internet. There are many benefits of online transactions like, fast transaction speed, convenience, low risk of theft etc.

Some of the popular online transaction websites are:

- 1) IRCTC(Indian Railway Catering and Tourism Corporation), an online portal for booking flight and train tickets.
- 2) Flipkart, an online shopping portal for buying consumer products. Flipkart offers multiple payment methods like credit card, debit card, net banking, e-gift voucher, and Cash on Delivery.
- 3) EBay, an online portal for buying and selling goods.
- 4) Redbus, an online portal for booking bus tickets.

Online shopping: Online shopping is a form of electronic commerce where customers can buy or sell goods over the Internet.

Online shopping could be useful in situations when:

- 1) A customer does not have sufficient time to visit stores.
- 2) Visiting a store is more expensive than purchasing a product online.
- 3) A product or service that is not available in the local market is available online.

Internet security: Internet security is a branch of computer security specifically related to the Internet. Its objective is to establish rules and measures to use against attacks over the Internet.

Best Practices for Security:

- 1) Use strong password and keep on changing after every 2 -3 weeks.
- 2) Use encryption software to protect your data from unauthorized users.
- 3) Never save your username or password on shared computers.
- 4) Do not share personal information.
- 5) Use updated antivirus and antispyware software.
- 6) Clear browser cookies frequently.
- 7) Keep the operating system and software applications up to date.
- 8) Install Firewall : Firewalls analyse the network traffic and determine if the traffic should be allowed or not.
- 9) Never install software from unknown sources.
- 10) Remove unwanted or unknown software applications: These might have got installed without your knowledge when you have visited some websites.
- 11) Clear Data Stored In Browsers : It is not advisable to leave the web browser store data(username, password, credit card detail) particularly on public or shared computers.

General guidelines for managing strong passwords are:

- Keep the length of the password at least 12-14 characters if permitted.
- Avoid keeping passwords based on repetition, dictionary words, letter or number sequences, usernames, relative or pet names, etc.
- Including numbers, and symbols in passwords if allowed.
- Use capital and lower-case letters.
- Avoid using the same password for multiple sites or purposes.
- Avoid using something that the public or workmates know you strongly like or dislike.

NOTE :

- www.strongpasswordgenerator.com help to generate strong password.
- Example of Online threats are Phishing, email spoofing, chat spoofing, etc.

- Combination of alphanumeric and special characters could be used for creating a password that is not so easy to crack.
- Programs such as keyloggers track and transmit every key that is pressed on a keyboard (keystrokes) to unauthorized users

MAINTAIN WORKPLACE SAFETY

Every organization must follow a standard set of safety rules and procedures. These rules must be stated and displayed clearly at important areas . All the employees must be given a demonstrations and training to follow safety rules .

Basic safety rules to follow at workplace – Fire safety, Falls and slips, Electrical safety, Use of first aid.

Timely repairs should be carried out by a competent person/ organization at workplace / home to avoid/minimize any hazards.

Basic Fire safety rules in an organization are:

Most of the time fire can be prevented using appropriate measures.

- Fire escape plans must be installed at proper levels
- Conduct regular drills
- Smoke alarms must be placed at proper intervals
- Keep workplace a no-smoking zone
- Maintenance of safety equipment must be taken care of regularly

Falls and Slips Safety rules:

- Keep the moving area clean and clutter free.
- Workplace must be proper ventilated receive light.
- Wear non slippery footwear.
- Floors must be clean and dry
- Oil spills, dust must be immediately cleaned.

ELECTRICAL SAFETY RULES:

Following are some of the measures we should take to keep ourselves safe from electricity hazards and accidents:-

- Electrical equipment used should be approved by a recognised organization.
- Workers should be trained to handle the electric equipment safely
- Damaged and hazardous electrical equipment should be immediately replaced. All old and workout and frayed switches and wires should be changed.
- Heat emanating equipment should be kept away from the electrical equipment.
- Take care that the outlets/ circuits should not be overloaded .
- Switch off and unplug the electrical appliances before cleaning or relocating them.

USE OF FIRST AID

First Aid is the immediate assistance provided to the injured to save life and minimize health loss till the proper medical aid/ facility is provided.

Every organization must have basic First Aid Kit. Kit must have all necessary items. Make sure to check for the expiry of medical related items.

Some rules of First Aid are :

- Assure the injured to remain calm and not to panic
- Keep them warm if they are under shock
- Do not move the victim in case of back/neck injury

Occupational hazard: An occupational hazard is the illness one may acquire due to his occupation. Some types of occupational hazards are : Physical hazards, chemical hazards, biological hazards, , behavioural hazards, radiological hazards, ergonomic hazards etc.

Accident: an accident is an unplanned event that may happen all of a sudden and may lead to unwanted or unprecedented results/outcomes.

Types of Accidents: Accidents may be of following types :

- Accidents at workplace : Slips and fall accidents, fire
- Industrial disease/illness
- Road traffic accidents
- Clinical Accidents
- Sports related accidents

Handling Accidents:

Accidents must be handled carefully. The accident should be handled compassionately without assigning blame on others.

- Every organization must follow SOP for accident handling
- Safety measures must be placed to prevent workplace accidents
- Immediately call the medical team for any injury
- Stay alert
- Pay attention to and follow emergency drills

EMERGENCY: Any unexpected situation that needs immediate attention and action is called emergency.

An emergency situation is one that:

- threatens the employees, customers or the public
- disrupts or shuts down the operations
- causes physical or environmental damage

TYPES OF EMERGENCY:

Various types of emergencies are there and there should be an emergency management plan to handle the situation of emergency. Some of the types of emergencies are as follows :

- Chemical spills
- Extreme heat waves
- Droughts
- Pandemics
- Terrorist attack
- Fire
- Floods
- Thunderstorms
- Leakage of some hazardous gas/ chemical

Some of the types of emergencies that require evacuation are:

- Fire
- Explosion
- Floods
- Earthquake
- Hurricane
- Tornado
- Toxic material release
- Civil disturbance
- Workplace violence

HAZARDS AND SOURCES OF HAZARDS:

A hazard is anything that is the source of any potential harm, damage or any kind of potential loss of health or life.

The different types of hazards include:

- Physical
- Chemical
- Biological
- Mechanical

Evacuation: Evacuation is the process of emptying a place in case of an emergency, disaster. Every company must ensure following points for evacuation in case of any emergency:

- An evacuation policy.
- Organization must have a designated assembly point for emergencies.
- Floor plans with evacuation routes pasted in work areas
- Periodic evacuation drills should be conducted

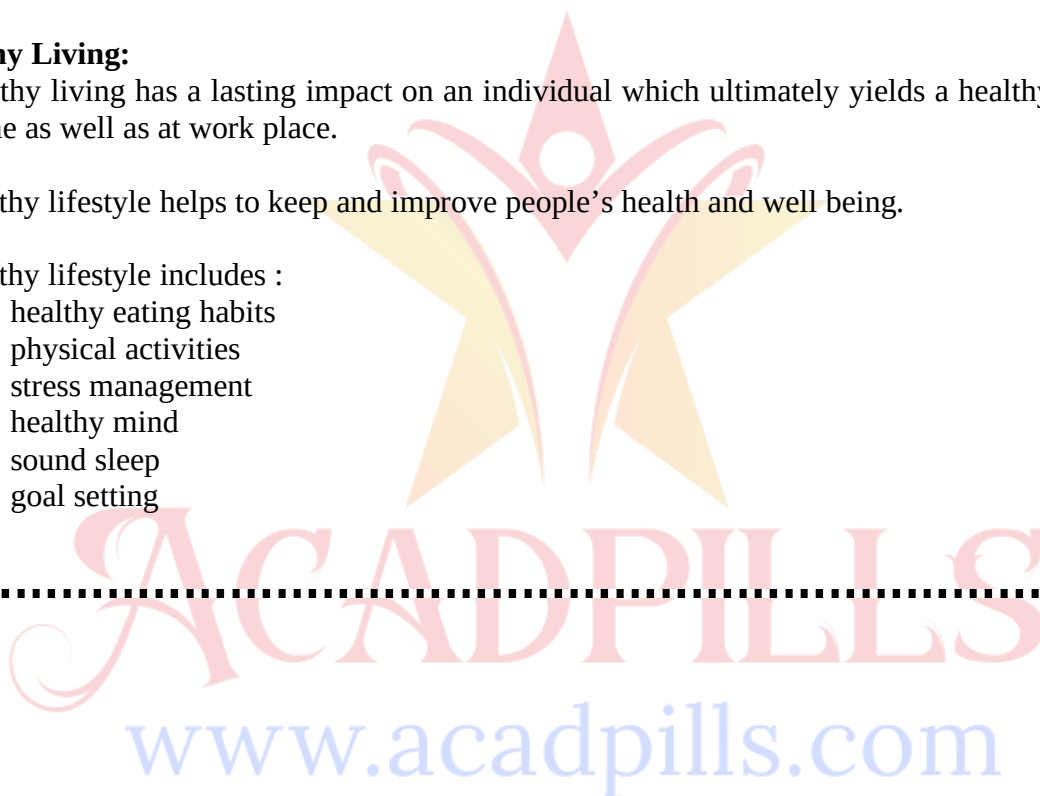
Healthy Living:

A healthy living has a lasting impact on an individual which ultimately yields a healthy environment at home as well as at work place.

A healthy lifestyle helps to keep and improve people's health and well being.

A healthy lifestyle includes :

- healthy eating habits
- physical activities
- stress management
- healthy mind
- sound sleep
- goal setting



WEB APPLICATIONS AND SECURITY

OBJECTIVE QUESTIONS (SET 01)

1. Computer _____ refers to the user friendliness of a computer system for all, regardless of their disability.

- a. Shortcut Keys
- b. Accessibility
- c. Ctrl & Alt Key
- d. None of the above

Ans: b. Accessibility

2. What are the different type of impairments that impact computer usage?

- a. Dyslexia
- b. Visual Impairment
- c. Hearing Impairment
- d. All of the above

Ans: d. All of the above

3. Accessibility options are available in _____.

- a. Setting
- b. Control Panel
- c. Desktop
- d. None of the above

Ans: b. Control Panel

4. _____ is an accessibility feature to help computer users with physical disabilities, but it is also used by others as a means to reduce repetitive strain.

- a. Filter keys
- b. Toggle Keys
- c. Sticky Keys
- d. None of the above

Ans: c. Sticky Keys

5. _____ enables the user to hold and release a modifier key, like as Shift, Ctrl, Alt, or the Windows key, and have it remain active until another key is pressed.

- a. Filter keys
- b. Toggle Keys
- c. Sticky Keys
- d. None of the above

Ans: c. Sticky Keys

6. The _____ accessibility function instructs the keyboard to ignore short or repeated inputs, making typing simpler for those people who have hand tremors.

- a. Filter keys
- b. Toggle Keys
- c. Sticky Keys
- d. None of the above

Ans: a. Filter keys

7. _____ is an accessibility feature for persons with vision impairment or cognitive problems.

- a. Filter keys

- b. Toggle Keys
 - c. Sticky Keys
 - d. None of the above
- Ans: b. Toggle Keys

8. _____ is turned on, the computer emits sound cues when the locking keys such as Caps lock, Num lock and Scroll lock are pressed.

- a. Filter keys
 - b. Toggle Keys
 - c. Sticky Keys
 - d. None of the above
- Ans: b. Toggle Keys

9. _____ is designed to help users with auditory impairments.

- a. Filter keys
 - b. Toggle Keys
 - c. Sound Sentry
 - d. None of the above
- Ans: c. Sound Sentry

10. When the computer makes a sound, _____ produces visual effects such as a blinking title bar or a flashing border.

- a. Filter keys
 - b. Toggle Keys
 - c. Sound Sentry
 - d. None of the above
- Ans: c. Sound Sentry

11. _____ directs applications that provide information through sound to also give information visually, through written captions or informative icons.

- a. Filter keys
 - b. Show Sound
 - c. Sound Sentry
 - d. None of the above
- Ans: b. Show Sound

12. _____ is an accessibility feature to assist people with vision impairment. You can change the text size and color, as well as the background color, to make it easier to read.

- a. High Contrast
 - b. Show Sound
 - c. Sound Sentry
 - d. None of the above
- Ans: a. High Contrast

13. _____ is also an accessibility feature that assists people with vision impairment by changing the blink rate and width of the cursor.

- a. High Contrast
 - b. Cursor Options
 - c. Sound Sentry
 - d. None of the above
- Ans: b. Cursor Options

14. The _____ option uses the keyboard (especially numeric keypad) as a pointing device instead of a mouse.

- a. High Contrast
- b. Mouse key
- c. Sound Sentry
- d. None of the above

Ans: b. Mouse key

15. _____ is an accessibility feature that assists people that have difficulty using a keyboard or a mouse (or both). They can use special devices such as Sip, Puff and Breath Switches to provide input to the computer through Serial Ports.

- a. High Contrast
- b. Mouse key
- c. Serial Keys
- d. None of the above

Ans: c. Serial Keys

16. The option in Microsoft Windows XP used for helping users with physical disabilities and to reduce repetitive strain is _____.

- a. Toggle Key
- b. Sticky Key
- c. Tab Key
- d. None of the above

Ans: b. Sticky Key

17. Sound Sentry is intended to assist users who have _____ disabilities.

- a. Auditory Impairments
 - b. Cognitive Impairments
 - c. Visual Impairments
 - d. Dexterity Impairments
- Ans: a. Auditory Impairments

18. The High Contrast feature in Microsoft Windows XP is designed to help persons who have _____ impairments.

- a. Auditory Impairments
- b. Cognitive Impairments
- c. Visual Impairments
- d. Dexterity Impairments

Ans: c. Visual Impairments

19. _____ is designed to assist people that have difficulty using a keyboard or a mouse.

- a. KeyboardKeys
- b. MouseKeys
- c. TabKeys
- d. GeneralKeys

Ans: b. MouseKeys

20. A _____ is a collection of computers and other hardware components interconnected by communication channels that allow sharing of resources and information.

- a. Computer Network
- b. Computer Sharing
- c. Computer Receiving

d. None of the above
Ans: a. Computer Network

21. What are the examples of communication channels _____.

- a. Cable Wire
 - b. Satellites
 - c. Both a) and b)
 - d. None of the above
- Ans: c. Both a) and b)

22. Example of computer networks architecture _____.

- a. Peer – to – Peer architecture
 - b. Client – Server architecture
 - c. Both a) and b)
 - d. None of the above
- Ans: c. Both a) and b)

23. Networks in which all computers have an equal status are called _____.

- a. Peer – to – Peer Network
 - b. Client – Server Network
 - c. Both a) and b)
 - d. None of the above
- Ans: a. Peer – to – Peer Network

24. Networks in which certain computers have special dedicated tasks, providing services to other computers are called _____.

- a. Peer – to – Peer Network
 - b. Client – Server Network
 - c. Both a) and b)
 - d. None of the above
- Ans: b. Client – Server Network

25. The computer provides services called _____.

- a. Client
 - b. Server
 - c. Node
 - d. None of the above
- Ans: b. Server

26. The computer uses the services which are provided from the server is called _____.

- a. Client
 - b. Server
 - c. Node
 - d. None of the above
- Ans: . Client

27. What are the major different types of network?

- a. Local Area Network
 - b. Wide Area Network
 - c. Both a) and b)
 - d. None of the above
- Ans: c. Both a) and b)

28. A _____ is a network that connects computers and gadgets in a specific geographic area, such as a house, school, computer laboratory, office building, or group of buildings.

- a. Local Area Network
- b. Wide Area Network
- c. Both a) and b)
- d. None of the above

Ans: a. Local Area Network

29. A _____ network is one which covers a broad area across metropolitan city, regional, or national boundaries.

- a. Local Area Network
- b. Wide Area Network
- c. Both a) and b)
- d. None of the above

Ans: b. Wide Area Network

30. The _____ is a global network of interconnected computer networks that serve billions of people worldwide by utilizing the standard Internet protocol suite.

- a. Internet
- b. Network
- c. Arpanet
- d. None of the above

Ans: a. Internet

31. WWW stands for _____.

- a. Web World Wide
- b. World Wide Web
- c. World Web Wide
- d. None of the above

Ans: b. World Wide Web

32. WWW is also known as _____.

- a. W2
- b. W4
- c. W3
- d. W1

Ans: c. W3

33. _____ is a network of interconnected hypertext documents that may be accessed via the Internet.

- a. World Wide Web
- b. W3
- c. Both a) and b)
- d. None of the above

Ans: c. Both a) and b)

34. A _____ is a software program that is used to browse Web sites and serves as a bridge between the user and the World Wide Web.

- a. World Wide Web
- b. Web Server
- c. Web Browser
- d. None of the above

Ans: c. Web Browser

35. What are the advantages associated with networking?

- a. Data & File Sharing
- b. Hardware Sharing
- c. Internet Sharing
- d. All of the above

Ans: d. All of the above

36. _____ helps to share hardware components such as printers, scanners etc.

- a. Computer Network
- b. Hardware Network
- c. Information Network
- d. None of the above

Ans: a. Computer Network

37. ISP stands for _____.

- a. Internet Sharing Provider
- b. Internet Services Provider
- c. Internet Social Provider
- d. All of the above

Ans: b. Internet Services Provider

38. Example of Internet Service Provider in India _____.

- a. Bharat Sanchar Nigam Limited
- b. Airtel
- c. MTS
- d. All of the above

Ans: d. All of the above

39. WiMAX stands for _____.

- a. WorldWide Interoperability for Microwave Access
- b. WorldWide Interoperability for Micro Access
- c. WorldWide Internet for Microwave Access
- d. None of the above

Ans: a. WorldWide Interoperability for Microwave Access

40. DSL stands for _____.

- a. Dynamic Subscriber Line
- b. Digital Subscriber Line
- c. Digital Sub Line
- d. None of the above

Ans: b. Digital Subscriber Line

41. ISDN stands for _____.

- a. Integrated Services Digital Network
- b. Internet Services Dynamic Network
- c. Integrated Sub Digital Network
- d. Internet Sub Dynamic Network

Ans: a. Integrated Services Digital Network

42. A _____ is a device that converts digital computer signals into analog signals that can travel over phone lines.

- a. Router

- b. Modem
- c. Bridge
- d. None of the above

Ans: b. Modem

43. Modem is also known as _____.

- a. MOdulator/ DEModulator
- b. Modular/DeModular
- c. Modulate/DeModulate
- d. None of the above

Ans: a. MOdulator/ DEModulator

44. Give the example of a Wired internet connectivity channel?

- a. DSL
- b. WiMax
- c. Wi-Fi
- c. All of the above

Ans: a. DSL

45. Give the example of a Wireless internet connectivity channel?

- a. DSL
- b. Dial – Up
- c. WiMax
- d. None of the above

Ans: c. WiMax

46. _____ Internet access is a form of Internet access that uses the facilities of the public switched telephone network (PSTN) to establish a connection to an Internet service provider

- a. Dial – up
- b. 3G
- c. WiMax
- d. Wi-Fi

Ans: a. Dial – up

47. _____ provide Internet access by transmitting digital data over wires of a local telephone network.

- a. Dial – Up
- b. Digital Subscriber Link (DSL)
- c. 3G
- d. WiMax

Ans: b. Digital Subscriber Link (DSL)

48. _____ provides wireless communication and mobile broadband connection across cities and countries.

- a. WiMax
- b. Wi-Fi
- c. DSL
- d. None of the above

Ans: a. WiMax

49. _____ is a widely used technology that allows electronic devices as as computers or mobile phones to transmit data wirelessly over a network, including high-speed Internet connections.

- a. WiMax

- b. Wi-Fi
- c. DSL
- d. None of the above

Ans: b. Wi-Fi

50. How can we transfer data on the internet?

- a. Using Box Switching
- b. Using Packet Switching
- c. Using Public Switching
- d. Using Web Switching

Ans: b. Using Packet Switching

51. The acronym for LAN is _____.

- a. Large Area Network
- b. Last Area Network
- c. Local Area Network
- d. None of the above

Ans: c. Local Area Network

52. Three types of Wired Internet Connectivity are _____, _____ & _____.

- a. 3G, DSL, Dial-Up
- b. Cable Internet Access, DSL, Dial-Up
- c. WiMax, 3G, Wi-Fi
- d. WiMax, DSL, Dial-up

Ans: b. Cable Internet Access, DSL, Dial-Up

53. Three types of Wireless Internet Connectivity are _____, _____ & _____.

- a. 3G, DSL, Dial-Up
- b. Cable Internet Access, DSL, Dial-Up
- c. WiMax, 3G, Wi-Fi
- d. WiMax, DSL, Dial-up

Ans: c. WiMax, 3G, Wi-Fi

54. _____ is a form of communication over the Internet that offers an instantaneous transmission of text-based messages from sender to receiver.

- a. Instant Messaging
- b. Computer Software
- c. Hardware
- d. None of the above

Ans: a. Instant Messaging

55. Most instant messaging software include the option for performing_____.

- a. File transfer
- b. Audio chat
- c. Video calling
- d. All of the above

Ans: d. All of the above

56. What are the key features of instant messaging _____.

- a. Text Message can be sent to one or more person
- b. Audio calling and conferencing
- c. Video calling and conferencing
- d. All of the above

Ans: d. All of the above

57. Give some examples of application based instant messaging software?

- a. Skype
- b. Yahoo! Messenger
- c. Rediff Bol
- d. All of the above

Ans: d. All of the above

58. Give some examples of web based instant messaging software?

- a. Meebo
- b. Yahoo! Messenger
- c. IMO
- d. All of the above

Ans: d. All of the above

59. _____ is a type of Internet communication software that allows the user for instant transfer of text-based communications from sender to receiver.

- a. Instant Messaging (IM)
 - b. Service Messaging (SM)
 - c. Fast Messaging (FM)
 - d. None of the above
- Ans: a. Instant Messaging (IM)

60. _____, _____ & _____ & Web Camera are required for audio and video conferencing.

- a. Joystick, Computer, Video
 - b. Text, Audio, Video
 - c. Microphone, Headphone, Speaker
 - d. Touch Screen, Speaker, Microphone
- Ans: c. Microphone, Headphone, Speaker

61. What are the general rules and etiquettes for chatting?

- a. Messages should be short and to the point
 - b. Always ask first if the other person has time to converse.
 - c. Typing your message in uppercase is extremely rude
 - d. All of the above
- Ans: d. All of the above

62. A _____ is a discussion-style website that non-technical (or technical) people use to create personal web sites.

- a. Blog
- b. Website
- c. Social Media Site
- d. None of the above

Ans: a. Blog

63. _____ are similar to an online personal diary and simple to use.

- a. Blog
- b. Website
- c. Social Media Site
- d. None of the above

Ans: a. Blog

64. What are the popular blogs websites on the internet _____.

- a. WordPress
- b. Blogger
- c. Blog
- d. All of the above

Ans: d. All of the above

65. Give an example of offline blog editors _____.

- a. Qumana
- b. Windows Live Writer
- c. Blogdesk
- d. All of the above

Ans: d. All of the above

66. _____ deals with transfer of money over the internet.

- a. Online transactions
- b. Offline transactions
- c. Both a) and b)
- d. None of the above

Ans: a. Online transactions

67. Which protocols are used to do the translation process securely?

- a. Online Transfer process
- b. Online Transaction Process
- c. Both a) and b)
- d. None of the above

Ans: b. Online Transaction Process

68. In what situation online shopping is useful.

- a. A customer does not have enough time to go shopping.
- b. Visiting a store expenses more than purchasing a product online.
- c. A product or service that is not available in the local market only available online
- d. All of the above

Ans: d. All of the above

69. Give some of the popular online transaction websites.

- a. IRCTC
- b. Flipkart
- c. EBay
- d. All of the above

Ans: d. All of the above

70. What are the different online threats _____.

- a. Phishing
- b. Email Spoofing
- c. Chat Spoofing
- d. All of the above

Ans: d. All of the above

71. How can you reduce the risks over the internet?

- a. Using Antivirus Software
- b. Fire walls

- c. Strong Password
 - d. All of the above
- Ans: d. All of the above

72. Which combination is required to create a strong password.

- a. Combination of Alphanumeric
- b. Combination of Numbers
- c. Combination of special characters
- d. All of the above

Ans: d. All of the above

73. Do not keep password names such as _____.

- a. Favorite color
- b. Name of your friends & Relatives
- c. Don't use mobile number as password
- d. All of the above

Ans: d. All of the above

74. Give the name of the website which can generate strong passwords automatically?

- a. www.strongpasswordgenerator.com
- b. www.wikipedia.com
- c. www.cbseacademic.nic.in
- d. None of the above

Ans: a. www.strongpasswordgenerator.com

75. _____ software helps to protect your data from unauthorized users.

- a. W33 Software
- b. Encryption Software
- c. Unlock password Software
- d. None of the above

Ans: b. Encryption Software

76. How we can keep our username and password private.

- a. Don't tell username & password to others
- b. Don't share username & password in internet café
- c. Always clear browser history & cookies after using
- d. All of the above

Ans: d. All of the above

77. How can I make secure transactions on the internet?

- a. Use Antivirus and Antispyware programs
- b. Your Firewall should be always ON
- c. Open only that website which have a https://
- d. All of the above

Ans: d. All of the above

78. _____ are programs that are created on your local computer when you visit websites.

- a. Firewall
- b. Cookies
- c. History
- d. All of the above

Ans: b. Cookies

79. _____ can be software or hardware that helps in the security of a computer and a network.

- a. Firewall
- b. Cookies
- c. History
- d. All of the above

Ans: a. Firewall

80. Web browsers have _____ designed to store passwords used in forms on websites.

- a. Cookies
- b. Built – in password management
- c. History
- d. None of the above

Ans: b. Built – in password management

81. What basic safety rules to follow at the workplace.

- a. Fire Safety & Use of First aid
- b. Falls and slips
- c. Electrical Safety
- d. All of the above

Ans: d. All of the above

82. Basic fire safety rules in an organization are _____.

- a. Fire escape plans must be installed at proper levels
- b. Conduct regular drills & Smoke alarms must be placed
- c. Maintenance of safety equipment and No – smoking zone
- d. All of the above

Ans: d. All of the above

83. Basic falls and slips safety rules in an organization are _____.

- a. Maintain a clean and clutter-free working area.
- b. Workplace should be ventilated and receive light
- c. Floors must be clean and dry
- d. All of the above

Ans: d. All of the above

84. What are the different electrical safety rules?

- a. Electrical equipment should be approved by a recognized organization
- b. Worker should be trained to handle the electric equipment safely
- c. Damaged and hazardous electrical equipment should be immediately replaced
- d. All of the above

Ans: d. All of the above

85. Basic rules of using first aid are _____.

- a. Injured person should calm not to panic
- b. Keep them warm if they are under shock
- c. In the case of a back/neck injury, do not move the victim.
- d. All of the above

Ans: d. All of the above

86. What are the different types of accidents?

- a. Accidents at workplace
- b. Industrial disease/illness
- c. Clinical Accident

d. All of the above
Ans: d. All of the above

87. What protection should organizations take to handle accidents in the workplace?

- a. Safety measures must be placed to prevent workplace accidents
- b. Immediately call the medical team for any injury
- c. Pay attention to and follow emergency drills
- d. All of the above

Ans: d. All of the above

88. What are the different types of hazards include _____.

- a. Physical
- b. Chemical
- c. Biological
- d. All of the above

Ans: d. All of the above

89. _____ is the process of emptying a place in case of an emergency, disaster.

- a. Attention
- b. Evacuation
- c. Taking care
- d. None of the above

Ans: b. Evacuation

90. What evacuation emergency should be followed by every company.

- a. Company should have a evacuation policy
- b. Organization must have a designated assembly point for emergencies
- c. Floor plans with evacuation routes should be pasted in every floor
- d. All of the above

Ans: d. All of the above

91. Give some examples of a healthy lifestyle.

- a. Healthy eating habits
- b. Physical activities
- c. Stress Management
- d. All of the above

Ans: d. All of the above

92. An organization can face some _____ that could put the lives of the employees in danger.

- a. Issue
- b. Health Hazards
- c. Shortage
- d. None of the above

Ans: b. Health Hazards

93. Hazards can be of different types depending on the _____ and the _____ in which the employees work.

- a. Industry, Environment
- b. Occupation, problem
- c. Both a) and b)
- d. None of the above

Ans: a. Industry, Environment

94. Organizations must have a designated _____ for emergencies.

- a. Evacuation policy
- b. Assembly point for emergencies
- c. Buddy system for special needs people
- d. All of the above

Ans: d. All of the above

OBJECTIVE QUESTIONS (SET 02)

Q1. Which portal is used for booking train tickets?

- a. Flipkart
- b. EBay
- c. IRCTC
- d. Redbus

Ans: c. IRCTC

Q2. COD (in reference to online shopping) stands for ____

- a. Cash Over Delivery
- b. Cash on Delivery
- c. Catch on Delivery
- d. None of the above

Ans: b. Cash on Delivery

Q3. Which of the following payment method will allow customer to pay after receiving the product.

- a. Debit Card
- b. Credit Card
- c. COD
- d. None of the above

Ans: c. COD

Q4. IRCTC stands for _____

- a. Indian Railway Catering and Tourism Coordinator
- b. Indian Railway Catering and Tourism Corporation
- c. Indian Railway Catering and Tourist Corporation
- d. None of the above

Ans: b. Indian Railway Catering and Tourism Corporation

Q5. Which of the following are Online threats?

- a. Phishing
- b. Email spoofing
- c. Chat spoofing
- d. All of the above

Ans: d. All of the above

Q6. Online threats can be reduced by using _____

- a. Antivirus
- b. Firewall
- c. Strong Password
- d. All of the above

Ans: d. All of the above

Q7. Which of the following passwords is not easy to crack?

- a. favorite color as password

- b. friends name as password
 - c. mobile number as password
 - d. combination of alphanumeric and special characters as password
- Ans: d. combination of alphanumeric and special characters as password

Q8. Using _____ password can lower the risk of a security breach

- a. Weak
- b. Strong
- c. Week
- d. None of the above

Ans: b. Strong

Q9. www.strongpasswordgenerator.com can generate random strong passwords.(T/F)

- a. True
- b. False

Ans: b. False

Q10. _____ could transmit every key pressed on a keyboard to unauthorized users.

- a. Keylog
- b. Keyloggers
- c. Keypress
- d. None of the above

Ans: b. Keyloggers

Q11. General rule to follow for online secure transactions are

- a. Do not share personal information
- b. Use antivirus and antispyware software
- c. Use only secure websites for such transactions
- d. All of the above

Ans: d. All of the above

Q12. Website uses secure transaction is indicated through a digital certificate represented as a _____ symbol in the web browser's address bar.

- a. Lock
- b. Key
- c. Both of the above
- d. None of the above

Ans: a. Lock

Q13. Cookies are _____ that are created on your local computer when you visit websites.

- a. small files
- b. folder
- c. virus
- d. None of the above

Ans: a. small files

Q14. Updated antivirus software helps in protecting your computer from recent threats.(T/F)

- a. True
- b. False

Ans: a. True

Q15. Firewalls could be a _____

- a. Software

- b. Hardware
 - c. Either of the above
 - d. None of the above
- Ans: c. Either of the above

- Q16. Best Practices of computer security are
- a. Clear browser cookies frequently
 - b. Keep the operating system and software applications up to date
 - c. Install firewalls
 - d. All of the above
- Ans: d. All of the above

- Q17. Examples of Web browsers are _____
- a. Edge
 - b. Mozilla Firefox
 - c. Opera
 - d. All of the above
- Ans: d. All of the above

- Q18. Basic safety rules to follow at workplace includes
- a. Fire safety
 - b. Falls and slips
 - c. Electrical safety
 - d. All of the above
- Ans: d. All of the above

- Q19. Basic Fire safety rules in an organization are :
- a. Conduct regular drills
 - b. Smoke alarms must be placed at proper intervals
 - c. Both of the above
 - d. None of the above
- Ans: c. Both of the above

- Q20. Keep workplace a no-smoking zone is a basic rule of _____
- a. Fire Safety
 - b. Falls and Slips
 - c. Electrical Safety
 - d. First Aid
- Ans: a. Fire Safety

- Q21. Which of the following are some of the measures to keep ourselves safe from electricity hazards and accidents?
- a. Electrical equipment used should be approved by a recognized organization.
 - b. Workers should be trained to handle the electric equipment safely
 - c. Circuits should not be overloaded .
 - d. All of the above
- Ans: d. All of the above

- Q22. Learning _____ is the social and civic responsibility of every human being.
- a. First Aid
 - b. Computer
 - c. Handling electrical equipment
 - d. None of the above

Ans: a. First Aid

Q23. It is our responsibility to check for the _____ of medical related items in First Aid kit.

- a. Cost
- b. Expiry
- c. Colour
- d. None of the above

Ans: b. Expiry

Q24. Rules of First Aid are :

- a. Assure the injured to remain calm and not to panic
- b. Do not move the victim in case of back/neck injury
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q25. Hazard which occur due to occupation is called _____

- a. Occupational hazard
- b. Chemical Hazard
- c. Electrical Hazard
- d. All of the above

Ans: a. Occupational hazard

Q26. Types of Occupational Hazards are:

- a. Physical hazards
- b. Biological hazards
- c. Ergonomic hazards
- d. All of the above

Ans: d. All of the above

Q27. Unfortunate incident that occurs unintentionally causing hazardous result or injury or unwanted results is called _____

- a. Emergency
- b. Hazard
- c. Accident
- d. None of the above

Ans: c. Accident

Q28. Examples of accidents at work place are :

- a. Slips and fall accidents
- b. Fire accidents
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q29. Any unexpected situation that needs immediate attention and action is called _____

- a. Accident
- b. Emergency
- c. Occupational Hazard
- d. None of the above

Ans: b. Emergency

Q30. Source of any potential harm is called _____

- a. Emergency
- b. Accident
- c. Hazard
- d. All of the above

Ans: c. Hazard

Q31. Which of the following are impairment that impact computer usage?

- a. Cognitive impairments and learning disabilities
- b. Visual impairment
- c. Motor or dexterity impairment
- d. All of the above

Ans: d. All of the above

Q32. ADHD stands for _____

- a. Attention Deficiency Hyperactivity Disorder
- b. Attention Deficit Hyperactivity Disorder
- c. Attention Deficit Hyper Disorder
- d. Attention Default Hyperactivity Disorder

Ans: b. Attention Deficit Hyperactivity Disorder

Q33. Accessibility Options in Control Panel are used to customize _____

- a. Keyboard
- b. Display
- c. Mouse
- d. All of the above

Ans: d. All of the above

Q34. Which of the following key help to ignore brief or repeated keystrokes, making typing easier for people with hand tremors.?

- a. Toggle Key
- b. Sticky Key
- c. Filter Key
- d. All of the above

Ans: c. Filter Key

Q35. When ToggleKeys is turned on, computer emits sound cues when the locking keys (Caps Lock, Num Lock, or Scroll Lock) are pressed. A high sound is emitted when the keys are switched _____ and a low sound is emitted when they are switched _____

- a. Off, On
- b. On, Off
- c. Off, Off
- d. On, On

Ans: b. On, Off

Q36. Use Soundsentry and use showsounds are options available in _____ tab of Accessibility Options window.

- a. Keyboard
- b. Display
- c. Sound
- d. General

Ans: c. Sound

Q37. Which of the following option is not available on Keyboard tab of Accessibility Options Window?

- a. Toggle keys
- b. Filter keys
- c. Sticky Keys
- d. Hand Keys

Ans: d. Hand Keys

Q38. Once the Filter key is active, it's icon appears in the _____

- a. System tray
- b. Status bar
- c. Title bar
- d. Menu bar

Ans: a. System tray

Q39. Cursor Options is an accessibility feature that assists people with _____ impairment

- a. Vision impairment
- b. Hearing impairment
- c. Cognitive impairments
- d. None of the above

Ans: a. Vision impairment

Q40. In Cursor option, we can change the _____ and _____ of the cursor.

- a. size and width
- b. blink rate, width
- c. height and width
- d. length and breadth

Ans: b. blink rate, width

Q41. Which of the following is not a networking architecture?

- a. Peer to Peer (P2P) Architecture
- b. Client Server Architecture
- c. Client to Peer architecture
- d. All of the above

Ans: c. Client to Peer architecture

Q42. In a skit based on "Computer Network", Amit is playing a role of a component who is providing services to other components or computers. Identify the role of Amit.

- a. Client
- b. Server
- c. Peer
- d. None of the above

Ans: b. Server

Q43. I am a global system of interconnected computer networks. I am a network of networks. Who am I?

- a. Intranet
- b. Internet
- c. LAN
- d. MAN

Ans: b. Internet

Q44. Internet is widely used by _____

- a. Students
- b. Business man
- c. Scientist
- d. All of the above

Ans: d. All of the above

Q45. Which of the following statement is not correct about Internet?

- a. The Internet is the largest encyclopedia for all age groups.
- b. Internet is a major source of entertainment for the general public.
- c. Internet helps in maintaining contact with friends and relatives.
- d. None of the above

Ans: d. None of the above

Q46. Which of the following software is required to view websites in computer?

- a. Web Server
- b. Web Browser
- c. Protocol
- d. Telnet

Ans: b. Web Browser

Q47. I am a software who acts as an interface between the user and the World Wide Web. Who am I?

- a. Protocol
- b. Web Server
- c. Web Browser
- d. Telnet

Ans: c. Web Browser

Q48. I am a document who may contain text, images, videos, other multimedia, and hyperlink. Who am I?

- a. Browser
- b. Web Page
- c. HTML
- d. WWW

Ans: b. Web Page

Q49. Web Pages stored on web servers are retrieved by using a _____

- a. Web browser
- b. Web Site
- c. WWW
- d. Database

Ans: a. Web browser

Q50. Which of the following is an advantage of Computer Networking?

- a. Hardware Sharing
- b. Internet Access Sharing
- c. Data Sharing
- d. All of the above

Ans: d. All of the above

Q51. I am a device which convert analog signal to digital signal and vice-versa. Who am I?

- a. Router
- b. Modem
- c. ISP

d. Firewall
Ans: b. Modem

Q52. The word modem is derived from its function _____

- a. MODulator/DEModulator
 - b. MODul/DEModul
 - c. MOdem/DEModem
 - d. MODEL/DEModel
- Ans: a. MODulator/DEModulator

Q53. Which of the following signal can travel over phone lines?

- a. Digital Signal
 - b. Analog Signal
 - c. Both of the above
 - d. None of the above
- Ans: b. Analog Signal

Q54. Which of the following signal can read by computer?

- a. Digital Signal
 - b. Analog Signal
 - c. Both of the above
 - d. None of the above
- Ans: a. Digital Signal

Q55. _____ is a form of Internet access that uses the facilities of the PSTN.

- a. DSL
 - b. WiMax
 - c. Dial-up
 - d. 3G
- Ans: c. Dial-up

Q56. For using a DSL connection, you need a _____ and a _____.

- a. DSL modem, subscription
 - b. Router, subscription
 - c. DSL modem, service
 - d. DSL modem, router
- Ans: a. DSL modem, subscription

Q57. _____ is provided through existing cable TV networks.

- a. Dial-up connections
 - b. Cable Internet Access
 - c. WiMAX
 - d. 3G
- Ans: b. Cable Internet Access

Q58. _____ is a long range system. It is a wireless communications standard designed to provide mobile broadband connectivity across cities.

- a. WiMAX
 - b. DSL
 - c. Cable Internet Access
 - d. Dial-up Connection
- Ans: a. WiMAX

Q59. Which technology allows an electronic device such as computers or mobile phones to exchange data wirelessly over a network?

- a. WiMax
- b. DSL
- c. Wi-Fi
- d. ISP

Ans: c. Wi-Fi

Q60. _____ is used where cables cannot be run (such as old buildings, outdoor areas) to provide network and Internet access.

- a. Cable Internet Access
- b. DSL
- c. Wi-Fi
- d. ISP

Ans: c. Wi-Fi

Q61. What type of network is formed by interconnected computer spread in different rooms of a building?

- a. MAN
- b. WAN
- c. PAN
- d. LAN

Ans: d. LAN

Q62. What type of network is formed by interconnected computer spread in across countries?

- a. MAN
- b. WAN
- c. PAN
- d. LAN

Ans: b. WAN

Q63. _____ convert analog signal to digital signal at _____ computer.

- a. Modem, Source
- b. Modem, Destination
- c. Router, Source
- d. Router, Destination

Ans: b. Modem, Destination

Q64. WiFi devices such as personal computer, smartphones etc. can connect to a network resource such as Internet through a device called the _____

- a. Web Server
- b. Wireless Access Point
- c. Web Page
- d. Web Site

Ans: b. Wireless Access Point

Q65. Data transferred on internet in the form of _____

- a. Piece
- b. Packets
- c. Protons
- d. None of the above

Ans: b. Packets

Q66. Instant messaging (IM) is a form of _____

- a. communication
- b. calculation
- c. internet
- d. none of the above

Ans: a. communication

Q67. Most instant messaging software include _____

- a. audio chat
- b. video calling
- c. conferencing
- d. All of the above

Ans: d. All of the above

Q68. Which of the following hardware is required for video conferencing?

- a. Microphone
- b. Speaker
- c. Web Camera
- d. All of the above

Ans: d. All of the above

Q69. Which of the following instant messaging software is not using email address for managing account?

- a. Yahoo! Messenger
- b. Windows Live Messenger
- c. Skype
- d. All of the above

Ans: c. Skype

Q70. _____ instant messaging software is downloaded and installed on user's computer.

- a. Application based
- b. Web based
- c. Both of the above
- d. None of the above

Ans: a. Application based

Q71. Which of the following happens in real-time and the response from participants can be spontaneous?

- a. Email
- b. Instant messaging
- c. Offline Messages
- d. None of the above

Ans: b. Instant messaging

Q72. Audio and Video calling are features of _____

- a. E-mail
- b. Instant messaging
- c. Offline messages
- d. All of the above

Ans: b. Instant messaging

Q73. Which of the following file types can be transferred through Instant messaging?

- a. Spread sheets

- b. Audio files
 - c. Video files
 - d. All of the above
- Ans: d. All of the above

Q74. Aman and Suman has to do project together. Both wants to share their ideas with each other. Aman's mom suggested _____ for real time interaction.

- a. Yahoo Messenger
- b. Facebook Messenger
- c. Windows Live Messenger
- d. All of the above

Ans: d. All of the above

Q75. Before starting Google Talk, a _____ account is required

- a. Gmail
- b. Yahoo
- c. Rediff
- d. Windows

Ans: a. Gmail

Q76. Whenever your friend in the contact list of Google Talk is online you can see the person along with a _____ dot

- a. Red
- b. Yellow
- c. Green
- d. Blue

Ans: c. Green

Q77. Etiquettes to be followed while chatting is _____

- a. Messages should be short and to the point.
- b. Always ask if the other person has time to chat first.
- c. Give people time to respond.
- d. All of the above

Ans: d. All of the above

Q78. Whenever we type our message in uppercase, it gives an impression of _____ to others.

- a. Fun
- b. Rude
- c. Simplicity
- d. Clarity

Ans: b. Rude

Q79. To start using Yahoo! Messenger, a _____ account is required.

- a. Yahoo Mail
- b. Gmail
- c. Outlook
- d. Microsoft

Ans: a. Yahoo Mail

Q80. To import your contacts in Yahoo Messenger from other sources such as Microsoft Outlook, Gmail, Hotmail, etc., you can use the _____ option.

- a. Export

- b. Import
 - c. Insert
 - d. Update
- Ans: b. Import

OBJECTIVE QUESTIONS (SET 03)

Q1. Computer Accessibility refers to the user friendliness of a computer system.(T/F)

- a. True
- b. False

Ans: a. True

Q2. A person with disability or impairment can not use computer. (T/F)

- a. True
- b. False

Ans: b. False

Q3. Computer Accessibility or friendliness of computer is known as _____

- a. Technology
- b. Assistance Technology
- c. Assistive Technology
- d. None of the above

Ans: c. Assistive Technology

Q4. Identify the cognitive impairment from the following.

- a. Dyslexia
- b. ADHD
- c. Autism
- d. All of the above

Ans: d. All of the above

Q5. Keyboard Accessibility options are useful for those which have _____

- a. Physical Disabilities
- b. Vision Impairment
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q6. In windows XP we can access accessibility options in _____

- a. Control Panel
- b. Windows
- c. Network
- d. MyComputer

Ans: a. Control Panel

Q7. Which of the following tab is not available in Accessibility options window (in Windows XP)

- a. Keyboard
- b. Display
- c. General
- d. Volume

Ans: d. Volume

Q8. _____ keys are used to reduce repetitive strain.

- a. Filter
- b. Toggle
- c. Sticky
- d. Mouse

Ans: c. Sticky

Q9. Which of the following is/are a modifier key?

- a. Alt
- b. Ctrl
- c. Windows
- d. All of the above

Ans: d. All of the above

Q10. Which of the following options allow us to press Ctrl key and 'A' key separately.

- a. Filter Keys
- b. Sticky Key
- c. Toggle Keys
- d. Mouse Keys

Ans: b. Sticky Key

Q11. Toggle key is a feature of _____

- a. Microsoft Office
- b. Microsoft Excel
- c. Microsoft Word
- d. Microsoft Windows

Ans: d. Microsoft Windows

Q12. Toggle Keys is designed for people who have _____

- a. Vision Impairment
- b. Cognitive Disabilities
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q13. Effect of toggle keys can be checked by pressing _____ key

- a. Num lock
- b. Caps lock
- c. Scroll Lock
- d. All of the above

Ans: d. All of the above

Q14. When toggle key is active, the locking keys when switched on produce _____ sound.

- a. high
- b. low
- c. no
- d. none of these

Ans: a. high

Q15. SoundSentry is designed to help users with _____ impairments.

- a. Visual
- b. Auditory
- c. both of the above
- d. none of the above

Ans: b. Auditory

Q16. Soundsentry generates visual warnings.(T/F)

- a. True
- b. False

Ans: a. True

Q17. High Contrast and Cursor Options are available under _____ tab.

- a. Keyboard
- b. Mouse
- c. Display
- d. Sound

Ans: c. Display

Q18. Number keys can be used to operate mouse when _____ is active.

- a. KeyboardKeys
- b. MouseKeys
- c. NumKeys
- d. None of the above.

Ans: b. MouseKeys

Q19. Serial keys are helpful for those people who have difficulty to use _____

- a. Keyboard
- b. Mouse
- c. All of the above
- d. None of the above

Ans: c. All of the above

Q20. Which of the following is not accessibility option?

- a. ToggleKeys
- b. MouseKeys
- c. SoundSentry
- d. Serial Port

Ans: d. Serial Port

Q21. A computer network allows sharing of resources and information.(T/F)

- a. True
- b. False

Ans: a. True

Q22. A computer network is a collection of 2 or _____ computers.

- a. 2
- b. 3
- c. 4
- d. more

Ans: d. more

Q23. P2P stands for _____

- a. Peer 2 Peer
- b. Peer To Peer
- c. Pipe To Pipe
- d. Pipe 2 Pipe

Ans: b. Peer To Peer

Q24. A Network in which all computers have an equal status is called _____

- a. Peer To Peer
- b. Peer 2 Peer
- c. LAN
- d. PtoP

Ans: a. Peer To Peer

Q25. The computer(s) which provide services are called _____

- a. Client
- b. Main Computer
- c. Server
- d. None of the Above

Ans: c. Server

Q26. Networks in which certain computers have special dedicated tasks, providing services to other computers (in the network) are called

- a. Client Network
- b. Server Topology
- c. Both of the above
- d. Client Server Network

Ans: d. Client Server Network

Q27. A computer who make a request for services in Client Server Architecture is called

- _____
- a. Client
 - b. Server
 - c. Both of the above
 - d. None of the above

Ans: a. Client

Q28. A _____ network is one which connects computers and devices in a limited geographical area such as home, school etc.

- a. MAN
- b. WAN
- c. LAN
- d. None of the above

Ans: c. LAN

Q29. Printer, Scanner can also be connected in Computer network.(T/F)

- a. True
- b. False

Ans: a. True

Q30. Internet is an example of _____

- a. LAN
- b. MAN
- c. WAN
- d. None of the above

Ans: c. WAN

Q31. Internet is _____

- a. network of networks

- b. International Network
 - c. used by businesses, governments, non-profit organizations etc
 - d. All of the above
- Ans: d. All of the above

Q32. WAN stands for _____

- a. Wide Area Network
- b. Wide Arial Network
- c. Wide Area Networking
- d. None of the above

Ans: a. Wide Area Network

Q33. In which network data travel faster?

- a. LAN
- b. WAN

Ans: a. LAN

Q34. Internet is used by

- a. Students
- b. Scientist
- c. Lecturer
- d. All of the above

Ans: d. All of the above

Q35. The _____ is a global system of interconnected computer networks.

- a. Intranet
- b. Internet
- c. ARPANET
- d. None of the above

Ans: b. Internet

Q36. WWW stands for _____

- a. World Wide Web master
- b. World Wider Web
- c. World Wide Web
- d. None of the above

Ans: c. World Wide Web

Q37. A web page may contain

- a. Images
- b. Videos
- c. Text
- d. All of the above

Ans: d. All of the above

Q38. A web page can be viewed with the help of _____ and _____

- a. web browser, active internet
- b. protocol, active internet
- c. protocol, web browser
- d. None of the above

Ans: a. web browser, active internet

Q39. Which of the following is not a web browser?

- a. Internet Explorer
- b. Mozilla Firefox
- c. Safari
- d. None of the above

Ans: d. None of the above

Q40. _____ is a computer that stores web sites and their related files for viewing on the Internet.

- a. Web Client
- b. Web
- c. Web Server
- d. WWW

Ans: c. Web Server

Q 41. Which of the following files can be shared over network?

- a. Text File
- b. Spreadsheet
- c. Audio File
- d. All of the above

Ans: d. All of the above

Q 42. Advantages of Networking are :

- a. Data Sharing
- b. File Transfer
- c. Hardware Sharing
- d. All of the above

Ans: d. All of the above

Q 43. Which of the following can be shared over network?

- a. Printer
- b. Internet
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q 44. ISP stands for _____

- a. Internet Service Provider
- b. Internet Service Pack
- c. Intranet Service Provider
- d. Internet Service Packet

Ans: a. Internet Service Provider

Q 45. BSNL stands for _____

- a. Bharat Sanchar Nigam Limited
- b. Bharat Samachar Nigam Limited
- c. Bihar Sanchar Nigam Limited
- d. None of the above

Ans: a. Bharat Sanchar Nigam Limited

Q 46. Common ISP are :

- a. Amazon
- b. Flipkart
- c. Both of the above
- d. None of the above

Ans: d. None of the above

Q 47. _____ convert the analog signal to digital and vice versa

- a. Switch
- b. Router
- c. Hub
- d. Modem

Ans: d. Modem

Q 48. MODEM stands for _____

- a. Model Demodel
- b. Modulator Demodulator
- c. Modern Demodulate
- d. None of the above

Ans: b. Modulator Demodulator

Q 49. Which of the following are the types of common internet connectivity.

- a. Dial – up
- b. DSL
- c. WiMAX
- d. All of the above

Ans: d. All of the above

Q 50. Which of the following is a wireless technology?

- a. 3G
- b. Dial-up
- c. DSL
- d. Cable Internet Access

Ans: a. 3G

Q 51. Dial up connection is the fastest internet connection.(T/F)

- a. True
- b. False

Ans: b. False

Q 52. _____ signals can travel over phone lines.

- a. Analog
- b. Digital
- c. Both of the above
- d. None of the above

Ans: a. Analog

Q 53. PSTN stands for _____

- a. Public Switched Telecom Network
- b. Private Switched Telephone Network
- c. Public Switched Telephone Network
- d. None of the above

Ans: c. Public Switched Telephone Network

Q 54. DSL stands for _____

- a. Digital Subscriber LAN
- b. Digital Subscriber Line
- c. Digital Subscriber telephone Line

d. None of the above
Ans: b. Digital Subscriber Line

Q 55. WiMAX is a _____ technology.

- a. Wireless
- b. Wired

Ans: a. Wireless

Q 56. Which of the following technology uses the cable TV infrastructure?

- a. WiMAX
- b. Dial up
- c. Cable Internet Access
- d. 3G

Ans: c. Cable Internet Access

Q 57. Which of the following technology is used for long distance?

- a. Dial up
- b. DSL
- c. Cable Internet Access
- d. WiMAX

Ans: d. WiMAX

Q 58. Which of the following technology is preferred in hilly terrain?

- a. WiMAX
- b. Cable Internet Access
- c. Dial up
- d. None of the Above

Ans: a. WiMAX

Q 59. WAP stands for _____

- a. Wireless Access Pointer
- b. Wireless Across Point
- c. Wireless Access Point
- d. None of the above

Ans: c. Wireless Access Point

Q 60. Which of the following technology is used to provide free internet in hotels, malls etc. to their customers.

- a. WiFi
- b. WiMAX
- c. Dial up
- d. Cable Internet Access

Ans: a. WiFi

Q 61. _____ is a form of communication over internet.

- a. Instant Messaging
- b. Email
- c. Video Call
- d. All of the above

Ans: d. All of the above

Q 62. _____ offers instantaneous transmission of text based messages from sender to receiver.

- a. Instant Messaging
- b. Email
- c. Traditional Post
- d. None of the above

Ans: a. Instant Messaging

Q 63. Most instant messaging software includes

- a. File transfer
- b. Video Calling
- c. Audio Chat
- d. All of the above

Ans: d. All of the above

Q 64. For audio and Video chat users should have _____

- a. Microphone
- b. Speakers
- c. Web Camera
- d. All of the above

Ans: d. All of the above

Q 65. Some instant messaging software allows users to view messages received when they are not logged on. These are called ____

- a. Online Messages
- b. Offline Messages
- c. Real Time Messages
- d. None of the above

Ans: b. Offline Messages

Q 66. Key Features of an instant messaging are:

- a. Text Messages can be sent to one or more person
- b. Audio calling and conferencing.
- c. Video calling and conferencing.
- d. All of the above

Ans: d. All of the above

Q 67. To use instant messaging software, a user must have a valid instant messaging account.(T/F)

- a. True
- b. False

Ans: a. True

Q 68. _____instant messaging software uses email address for managing account.

- a. Yahoo Messenger
- b. Whats App
- c. Skype
- d. None of the above

Ans: a. Yahoo Messenger

Q 69. Which of the following is Application based Instant Messaging Software?

- a. Google Talk
- b. Skype
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q 70. Which of the following Instant Messaging software should be installed before using it?

- a. IMO
- b. Meebo
- c. Google Talk
- d. All of the above

Ans: c. Google Talk

Q 71. Google Talk is a _____ instant messaging service.

- a. Free
- b. Paid
- c. None of the above

Ans: a. Free

Q 72. For using Google Talk, you should have _____ account.

- a. Yahoo
- b. Gmail
- c. Skype
- d. None of the above

Ans: b. Gmail

Q 73. Google Talk provides _____ communication.

- a. Text
- b. Voice
- c. Both Text and Voice
- d. None of the above

Ans: c. Both Text and Voice

Q 74. When ever any one in the contact list of Google Talk is online, you can see the person along with _____ dot

- a. Red
- b. Blue
- c. Yellow
- d. Green

Ans: d. Green

Q 75. Typing messages in Upper case is considered as rude behavior.(T/F)

- a. True
- b. False

Ans: a. True

Q 76. Choose the general etiquette's to be followed while chatting.

- a. Messages should be short and to the point.
- b. Give people time to respond.
- c. It's important to properly end an IM conversation
- d. All of the above

Ans: d. All of the above

Q 77. Before you start using Yahoo! Messenger, a _____ account is required

- a. Yahoo Mail
- b. Gmail
- c. Both of the above
- d. Neither of the above

Ans: a. Yahoo Mail

Q 78. Which of the following provide chat services?

- a. MSN
- b. Rediff
- c. Sify
- d. All of the above

Ans: d. All of the above

Q 79. Some instant messaging software such as Yahoo! Messenger, Windows Live Messenger use _____ for managing the account

- a. Standard Names
- b. Email addresses
- c. Mobile Number
- d. None of the above

Ans: b. Email addresses

Q 80. Blogs are similar to an online _____

- a. Book
- b. Notebook
- c. Diary
- d. None of the above

Ans: c. Diary

Q 81. We can use blogs to convey messages about _____

- a. events
- b. news
- c. reviews
- d. All of the above

Ans: d. All of the above

Q 82. Websites that offer free blog services are

- a. www.WordPress.com
- b. www.blogger.com
- c. www.blogsome.com
- d. All of the above

Ans: d. All of the above

Q 83. In WordPress we can create _____

- a. Website
- b. Blog
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q 84. Blog address should be unique.(T/F)

- a. True
- b. False

Ans: a. True

Q 85. Title is coming at the _____ of the blog.

- a. Top
- b. Bottom

- c. In between
- d. None of the above

Ans: a. Top

Q 86. A blog is a chronological display of content(T/F)

- a. True
- b. False

Ans: a. True

Q 87. Themes in WordPress are _____

- a. tools to change layout of blog
- b. tools to change design of blog
- c. both of the above
- d. None of the above

Ans: c. both of the above

Q 88. Others can view the blog post only after _____

- a. Publishing the post
- b. Creating the post
- c. Inserting pictures
- d. Inserting videos

Ans: a. Publishing the post

Q 89. We can create blog offline using _____

- a. any Offline Blog Editor
- b. Qumana
- c. Either of the above
- d. None of the above

Ans: c. Either of the above

Q 90. Which of the following is an offline blog editor?

- a. Qumana
- b. Windows Live Writer
- c. Blogdesk
- d. All of the above

Ans: d. All of the above

Q 91. We need an active internet connection to publish the post.(T/F)

- a. True
- b. False

Ans: a. True

Q 92. Qumana is _____ blog editor.

- a. free
- b. paid
- c. shareware
- d. None of the above

Ans: a. free

Q 93. Qumana is _____

- a. free
- b. easy to use
- c. simple

d. All of the above
Ans: d. All of the above

Q 94. Transfer of money on internet is called _____

- a. Online transaction
- b. Offline transaction
- c. None of the above
- d. Both of the above

Ans: a. Online transaction

Q 95. OLTP stands for _____

- a. Offline Transaction Process
- b. Online Transaction Process
- c. Online Transaction Procedure
- d. None of the above

Ans: b. Online Transaction Process

Q 96. Examples of online funds transfer are _____

- a. NEFT
- b. RTGS
- c. Both of the above
- d. None of the above

Ans: c. Both of the above

Q 97. Benefits of online transaction is/are

- a. Fast
- b. Convenient
- c. Low risk of theft
- d. All of the above

Ans: d. All of the above

Q 98. Online shopping is an example of _____

- a. E-commerce
- b. E-Booking
- c. E-reservation
- d. None of the above

Ans: a. E-commerce

Q 99. Customers can pay online using

- a. Credit Card
- b. Debit Card
- c. Internet Banking
- d. All of the above

Ans: d. All of the above

Q 100. A customer is doing Online shopping when

- a. A Customer has sufficient time to visit store
- b. A product is available offline at lower rates
- c. Visiting store is more expensive than purchasing product online
- d. None of the above

Ans: c. Visiting store is more expensive than purchasing product online

A. Fill in the blanks:

1. The option in Microsoft Windows XP used for helping users with physical disabilities and to reduce repetitive strain is **Sticky Keys**.
2. Sound Sentry is designed to help users with **auditory** impairments.
3. The High Contrast option in Microsoft Windows XP is designed to assist people with **Vision** impairments.
4. **Serial Keys** is designed to assist people that have difficulty using a keyboard or a mouse.
5. The acronym for LAN is **Local Area Network**.
6. Three types of Wired Internet Connectivity are **Dial up**, **DSL** & **Cable Internet Access**.
7. **Instant Messaging** is a form of communication over the Internet that offers an instantaneous transmission of text-based messages from sender to receiver.
8. **Microphone**, **Headsets** & **Speakers & Web Camera** are required for audio and video conferencing.
9. An organization can face some **Health Hazards** that could put the lives of the employees in danger.
10. Hazards can be of different types depending on the **Industry and the Environment** in which the employees work.

B. Subjective Questions:

1. Define networking?
Ans. A computer network is a collection of computers that allow sharing of resources and information.
2. Give any three advantages of networking.
Ans. Three advantages of networking are :
 1. Data Sharing:
 2. Files Transfer
 3. Hardware Sharing
 4. Internet Access Sharing
3. Explain the term packet with reference to data transfer.
Ans. Data transfer over the network is divided into small segments called packet. Each packet is sent from one computer to another computer until it finds its destination.
4. Explain Wifi and its significance.
Ans. Wi-Fi stands for wireless fidelity. Wi-Fi is a popular technology that allows an electronic device such as computers or mobile phones to exchange data wirelessly over a network. Wi-Fi is used where cables cannot be run (such as old buildings, outdoor areas) to provide network and Internet access
5. Compare LAN with WAN.
Ans.

LAN	WAN
It Stands for Local Area Network	It Stands for Wide Area Network
It covers a small geographical area.	It spreads across countries.

6. Expand the Following terms :
 - a. DSL : Digital Subscriber Line
 - b. ISP : Internet Service Provider

- c. Modem : Modulator and Demodulator
- d. WWW : World Wide Web
- e. LAN : Local Area Network
- f. MAN : Metropolitan Area Network
- g. WAN : Wide Area Network
- h. P2P : Peer to Peer

7. Explain P2P Architecture and how it is different from Client Server Architecture?

Ans. Networks in which all computers have an equal status are called peer to peer networks. Generally in such a network each terminal has an equally competent CPU.

Networks in which certain computers have special dedicated tasks, providing services to other computers (in the network) are called client server networks. The computer(s) which provide services are called servers and the ones that use these services are called clients.

8. List any five application based instant messaging software.

Ans. Five application based instant messaging software are :

- Google Talk
- Yahoo! Messenger
- Skype
- Windows Live Messenger
- Rediff Bol

9. What do you mean by instant messages?

Ans. Instant messaging (IM) is a form of communication over the Internet that offers an instantaneous transmission of text-based messages from sender to receiver.

10. Give any three key features of Instant Messaging.

Ans. Three key features of Instant Messaging are :

- Text Messages can be sent to one or more person
- Audio calling and conferencing.
- Video calling and conferencing.

11. State any 03 rules and etiquettes to be followed while chatting on the Internet.

Ans. Three rules and etiquettes to be followed while chatting on the Internet are :

- Messages should be short and to the point.
- Always introduce yourself by name if your screen name doesn't reflect it.
- Always ask if the other person has time to chat first.

12. What are the basic needs to use instant messaging (chat) software.

Ans. The basic needs to use instant messaging (chat) software are :

- A valid instant messaging account.
- An active internet Connection.
- A computer and a browser

13. What is a blog? Explain its use.

Ans. A blog is a discussion style site used by non-technical (or technical users) users for creating personal web pages. Blogs are similar to an online personal diary and simple to use.

We can use a blog to convey messages about events, announcements, news, reviews, etc.

14. List any 5 websites that provide blog service.

Ans. Five websites that provide blog service are :

- www.wordpress.com

- www.blogger.com
- www.blog.com
- www.weebly.com
- www.blogsome.com

15. Difference between web page and website.

Ans. A single HTML document is called a web page. A collection of web pages is called web site.

16. What do you mean by publishing a post?

Ans. Publishing a post means to make the post public. Once we finished typing the content, we need to publish the post for others to see. To view the post, you can type the blog address in the address bar of the web browser.

17. Explain the purpose of an offline blog editor.

Ans. When ever we do not have an active internet connection, we can create blogs using a blog application and publish the blog whenever internet connectivity is available.

18. List any five offline blog editors.

Ans. Five Offline blog editors are :

- Qumana
- Windows Live Writer
- Blogdesk
- MarsEdit
- BlogJet

19. Explain the purpose of Online transactions.

Ans. Online transactions have made transactions very convenient and simplified the workflow in many forms in a business. Online transactions deals with transfer of money over the internet.

20. List any five websites that allow online transactions.

Ans. Five websites that allow online transactions are :

- IRCTC
- Flipkart
- EBay
- RedBus
- Amazon

21. List any three payment tools to use online transactions.

Ans. Three payment tools to use online transactions are :

- Credit card
- Debit Card
- UPI
- Net Banking
- Paytm
- PhonePe

22. Give any two benefits of online transactions.

Ans. Two benefits of online transactions are :

- Money can be transferred immediately.
- It helps to reduce black marketing to some extent.

23. Explain the purpose of Internet Security.

Ans. Internet security is a branch of computer security specifically related to the Internet. Its objective is to establish rules and measures to use against attacks over the Internet.

24. Explain different kinds of online threats.

Ans. Different online threats are :

1. Phishing
2. Email spoofing
3. Chat spoofing

25. Explain the need to clear data stored in browsers.

Ans. Browsers often prompt to save usernames and passwords when users attempt to logon to websites. Browsers can also store data such as cookies, visited websites or webpages data, browsing history, etc. However it is not advisable to leave the web browser store this data particularly on public or shared computers. So it is advisable to clear data stored in browsers particularly on public or shared computers.

26. Explain any five tips to manage strong passwords.

Ans. Following are general guidelines for managing strong passwords :

- Keep the length of the password at least 12-14 characters if permitted.
- Avoid keeping passwords based on repetition, dictionary words, letter or number sequences, usernames, relative or pet names, etc.
- Including numbers, and symbols in passwords if allowed.
- Use capital and lower-case letters.
- Avoid using the same password for multiple sites or purposes.

27. Explain any four best practices to secure data.

Ans. Four best practices to secure data are :

- Keeping your username and password private.
- Use updated antivirus and antispyware software.
- Clear browser cookies frequently.
- Do not click on link shared by unknown users.
- Never install software from unknown sources.
- Install firewalls

28. Explain use of Antivirus and Anti Spyware.

Ans. Antivirus and Antispyware programs offer real-time protection monitoring your computer for any changes by malware software. Keep your Antivirus and Antispyware software always up to date, this can help in protecting your computer from any threats.

29. Enlist any three basic safety rules to follow at workplace.

Ans. Basic safety rules to follow at workplace –

1. Fire safety
2. Falls and slips
3. Electrical safety.

30. Give any two basic safety rules for ensuring Falls and Slips safety.

Ans. Two basic safety rules for ensuring Falls and Slips safety are as follows :

- a. Keep the moving area clean and clutter free.
- b. Workplace must be proper ventilated receive light.
- c. Wear non slippery footwear.
- d. Floors must be clean and dry.

31. Give any two electrical safety rules in any organization.

Ans. Two electrical safety rules in any organization are as follows :

1. Electrical equipment used should be approved by a recognized organization.
2. Workers should be trained to handle the electric equipment safely.

32. Explain any two first aid rules.

Ans. Two first aid rules are :

1. Assure the injured to remain calm and not to panic.
2. Keep them warm if they are under shock.

33. What do you mean by occupational hazards?

Ans. An occupational hazard is the illness one may acquire due to his occupation. for example people working in a chemical factory may get affected due to presence of certain chemicals.

34. List any three types of occupational hazards.

Ans. Three types of occupational hazards are as follows :

1. Physical hazards
2. Chemical hazards
3. Biological hazards

35. Explain the terms accident and emergency.

Ans. Accident can be defined as an unfortunate incident that occurs unintentionally causing hazardous result or injury or unwanted results.

Any unexpected situation that needs immediate attention and action is called emergency.

36. Enlist any four types of accidents.

Ans. Accidents may be of following types :

1. Accidents at workplace : Slips and fall accidents
2. Industrial disease/illness
3. Road traffic accidents
4. Clinical Accidents
5. Sports related accidents

37. Give any three situations of emergency that require evacuation.

Ans. Some of the types of emergencies that require evacuation are:

- Fire
- Explosion
- Floods
- Earthquake
- Hurricane
- Tornado
- Toxic material release
- Civil disturbance
- Workplace violence

38. Give any two ways to handle accidents.

Ans. Accidents must be handled carefully. Two ways to handle accidents are : (Write any two)

1. Every organization must follow SOP for accident handling.
2. Safety measures must be placed to prevent workplace accidents.
3. Immediately call the medical team for any injury.
4. Stay Alert

39. List any three types of hazards.

Ans. Different types of hazards are as follows : (Write any three)

Physical
Chemical

40. Give a checklist for workstations to minimize the hazards.

Ans. Checklist for Workstations : The workstation should:

1. Provide sufficient space for the user to alter position comfortably.
2. Provide adequate lighting.
3. Have windows fitted with adjustable coverings to alter the sunlight level.
4. Be spacious enough when a workstation is shared by more than one person.

41. Explain the term 'Evacuation Policy'.

Ans. Evacuation Policy means a plan to evacuate a place in case of any emergency or disaster. Evacuation is the process of emptying a place in case of an emergency, disaster. Every organization must have an evacuation policy.

42. Explain Buddy System to implement evacuation efficiently in case of emergency.

Ans. This system ensures that differently-abled are assisted and guided out of the premises or the impacted area properly. If you are a buddy to someone, ensure that your buddy is safe at the assembly point with you in case of any emergency.

43. Explain the terms Floor Plans and Assembly /areas.

Ans. Floor Plans means to show evacuation routes in work area. It is the responsibility of all employees or workers to understand these plans so that they can use in case of any emergency.

Assembly areas : A space where all employees or workers are required to assemble after evacuation.

44. Describe the importance of periodic evacuation drills.

Ans. Evacuation drills are very important for all employees or workers so that they know which path they have to follow to evacuate their floor/room/office in case of any emergency.

45. Explain importance of a healthy lifestyle.

Ans. Healthy Lifestyle leads to a healthy being. A healthy living has a lasting impact on an individual at home as well as at work place. A happy and healthy worker will always perform best to his ability.

46. Give any four points of a healthy lifestyle.

Ans. Healthy lifestyle includes : (Write any four)

1. Healthy eating habits
2. Physical activities
3. Stress management
4. Healthy mind
5. Sound sleep
6. Goal setting

SHORT ANSWER TYPE QUESTIONS

1. Name the term defined by given below statement: "A group of computers connected to each other by a link."

Ans. Computer network is defined as a group of computers connected to each other by a link that allows sharing of resources and information.

2. What are the main types of computer networks?

Ans. Following are the three main types of computer networks, based upon the geographical area as follows:

- (i) Local Area Network (LAN)
- (ii) Metropolitan Area Network (MAN)
- (iii) Wide Area Network (WAN)

3. What is the main purpose of switch?

Ans. A switch is a hardware device which is used to connect devices or segment of network with smaller subsets of LAN segments.

4. Identify the following devices.

- (i) A device that is used to connect different types of networks. It performs the necessary translation, so that the connected networks can communicate properly.
- (ii) A device that converts data from digital bit stream into an analog signal and vice-versa.

Ans. (i) Router (ii) Modem

5. What is the advantage of mesh topology?

Ans. In mesh topology, each node is connected to more than one node, so that it provides alternative route. In case, if the host is either down or busy.

6. Give the names of models of computer networking.

Ans. There are mainly two models of computer networking as follows:

- (i) Peer-to-Peer network (ii) Client-server network

7. How many techniques are used for transmitting data across network?

Ans. There are three techniques used for transmitting data across network : circuit switching, packet switching and message switching.

8. Write two advantages of using Internet.

Ans. Advantages of using Internet are as follows:

- (i) It is used for communication, entertainment, searching information and for providing many types of services.
- (ii) It provides the facility of E-mail.

9. Name any five DSL broadband service providers in India.

Ans. BSNL, Airtel, Reliance, MTNL and Tata Indicom are the five DSL broadband service providers in India.

10. Write the name of any two Internet Service Provider (ISP) in India.

Ans. MTNL and Airtel are the two Internet Service Provider (ISP) in India.

11. Name any two ways of wireless connections of Internet. Ramya wants to connect all the computers of her office wirelessly in order to avoid uncoordinated cables. Which wireless technology would be best suitable for her office?

Ans. Wi-Fi and WiMax are the two ways of wireless connections of Internet. Wi-Fi technology would be best suitable for her office to avoid cabling. Because, it allows high speed Internet connections without the use of cables or wires.

12. Write the full form of following terms.

(i) WiMAX

(ii) BPL

Ans. (i) WiMAX Worldwide Interoperability for Microwave Access.

(ii) BPL Broadband over Power Line

- 13.** What can a user do with WWW? Name any two major services provided by Internet.
Ans. Using WWW (World Wide Web), a user can download files, listen to music, view video files and jump to other documents or websites by using hypertext links. E-mail and WWW are the two major services provided by Internet.
- 14.** Mr. Lal owns a factory which manufactures automobile spare parts. Suggest him the advantages of having a web page for his factory.
Ans. The web page provides the information to the clients about his factory of spare parts. Moreover, he can receive the order on the Internet from the clients using the web page.
- 15.** Name two web browsers of Internet.
Ans. Internet Explorer and Google Chrome are the two web browsers of Internet.
- 16.** What is URL? Can we use URL to access a web page? How?
Ans. URL means Uniform Resource Locator. It is a unique address of a web page on the Internet. It specifies the Internet address of a file stored on a host computer connected to the Internet. Yes, as a location on a web server, which is called a website and each website has a unique address known as URL. So, an URL can be used to access a web page.
- 17.** Identify web addresses and E-mail addresses from the following.
(i) www.scrapbook.com
(ii) aba@scrapbook.com
(iii) www.countrywide.co.in
(iv) 123@hotmail.co.in
Ans. Key features of an Instant Messaging are:
(i) and (iii) are web addresses.
(ii) and (iv) are E-mail addresses.
- 18.** Write any two key features of an Instant Messaging (IM).
Ans. Key features of an Instant Messaging are :
(i) Video calling
(ii) Language translation
- 19.** Who is blogger? List any five websites that provide blog service.
Ans. A person who writes a blog or a weblog is known as blogger. Blogging is the act of posting content on a blog. Any five websites that provide blog service listed below
(i) Joomla (ii) Blogger (iii) Word Press.com (iv) Tumblr (v) Drupal
- 20.** Explain the purpose of a blog.
Ans. The main purpose of blog is to convey messages about events, announcements, need, review, etc. Blogs are usually managed using a web browser and this requires active internet connection.
- 21.** Laluma Chakradhar wants a broadband connection to access her mails and stay informed about the latest happening in the field of Biotechnology. Can you suggest two Internet Service Providers (ISPs) of India to be approached for the same?
Ans. BSNL and Airtel
- 22.** Explain the concept 'Watch Your waist and weight; to maintain a healthy body at workplace'.
Ans. The explanation of the concept 'Watch Your waist and weight; to maintain a healthy body at workplace' is

- (i) Encourage the use of weighing scales, measuring tapes and devices to monitor individual physical fitness.
- (ii) Place weighing scales, measurement tapes and body mass index charts in strategic locations.

23. What are the advantages of having safer and healthy workplace?

Ans. The advantages of having safer and healthy workplaces are as follows:

- (i) They have fewer days lost, therefore they have higher productivity.
- (ii) They can increase efficiency and quality, because they have healthier workplace.

24. What are physical hazards?

Ans. Physical hazards can be any factors within the environment that can harm the body without necessarily touching it. It can be caused by radiation (microwave, radio wave), pressure extremes (high pressure or vacuum), constant loud noise, high exposure to sunlight/ultraviolet rays, temperature extremes (hot and cold), etc.

25. What are slipping and tripping hazards?

Ans. Slipping hazards are caused through spills or trips such as cords running across the floor or water spilled on the floor. Tripping hazards are caused through falls such as working from heights, including ladders, scaffolds, roofs, etc.

26. How does water and foam fire extinguisher work?

Ans. Water and foam fire extinguishers extinguish the fire by taking the heat element away from the fire. Water extinguishers are for class A fires. Class A fires involve ordinary combustible materials, such as cloth, wood, paper, rubber and many plastics.

27. How does carbon dioxide fire extinguisher work?

Ans. Carbon Dioxide fire extinguishers extinguish fire by taking away the oxygen element of the fire triangle and also by removing the heat with a very cold discharge. Carbon dioxide can be used on Class B and C fires. They are usually ineffective on Class A fires.

28. How does wet chemical fire extinguisher work?

Ans. Wet chemical is a new agent that extinguishes the fire by removing the heat of the fire triangle and prevents re-ignition by creating a barrier between the oxygen and fuel elements. Wet chemical of Class K extinguishers were developed for modern, high efficiency deep fat fryers in commercial cooking operations. Some may also be used on Class A fires in commercial kitchens.

29. Write down any three points of differences between LAN, MAN and WAN.

Ans. Three major points of differences among LAN, MAN and WAN are as follows

- (i) In terms of geographical area WAN covers the area across the continents, MAN covers area within a city while LAN covers area within a building.
- (ii) In terms of distance The approximate distance covered by WAN is unlimited, of MAN is upto 160 km while that of LAN is upto 5 km.
- (iii) In terms of speed The speed with which the data travels in case of WAN is 10 to 38.4 Kbps, 183 to 7.2 Mbps, while that of MAN is 11 to 100 Mbps and of LAN is 11 to 54 Mbps.

30. Explain the LAN and WAN.

Ans. A Local Area Network (LAN) is a computer network covering a small geographical area like a home, office or small group of buildings such as buildings in a school. Computers connected to a LAN can share information and peripheral equipments.

A Wide Area Network (WAN) is used to connect LANs and other types of network together, so that users and computers in one location can communicate with users and computers in other locations.

31. What is the use of repeater in a network?

Ans. In a network, repeater receives a weak signal coming from a source and amplifies it to its original strength and then again forwards it to the destination. Basically, repeaters are used in a network so that a signal can travel longer distance.

32. When the computer network uses telephone lines as communication channel, then MODEM is used as a data communication device. Now, explain the working of MODEM.

Ans. MODEM performs the task of modulation at sender's site and demodulation at the receiver's site. Basically, our computer generates data in the form of digital signals which need to be forwarded to the receiver through telephone lines. Since, telephone lines can carry only analog signals. So, digital signals need to be converted to analog signals at sender's site this is called modulation. At receiver's site, again analog signals should be converted back to the original digital signals, then this is called demodulation.

33. Elucidate the following terms.

(i) PAN (ii) WiMAX (iii) Satellite

Ans. (i) PAN It stands for Personal Area Network. It is a computer network used for communication among computer and different technological devices close to it.

(ii) WiMAX It stands for Worldwide Interoperability for Microwave Access. It is a wireless transmission of data using a variety of transmission modes.

(iii) Satellite which are orbiting around the Earth, provide necessary links for telephone and television service.

34. Why do we use Instant Messaging?

Ans. Instant messaging is a set of communication technologies used for text based communication between two or more participants over the Internet or other types of networks. Many instant messaging services offer video calling features, Voice over Internet Protocol (VoIP) and web conferencing services. Web conferencing services can integrate both video calling and instant messaging abilities.

35. How can you prepare for emergencies?

Ans. To prepare for emergencies, follow the given guidelines

(i) Follow workspace protocol and guidelines to ensure safety and health; know and understand rules and procedures concerning fire emergencies.

(ii) Ensure that smoke alarms and sprinkler systems are installed, working properly and are not blocked.

(iii) Conduct regular fire drills.

36. Explain any three rules that companies follows in terms of fire safety.

Ans. Any four rules that companies follows in terms of fire safety are as follows

(i) All fire alarms systems are working.

(ii) All fire escapes can be opened easily.

(iii) Automatic fire doors close correctly.

37. What are biological hazards?

Ans. Biological hazards are caused by bacteria, viruses, insects, plants, birds, animals and humans, etc. It includes exposure to harm or disease associated with working with animals, people or infectious plant materials. Workplaces like schools, day care, colleges, hospitals, laboratories or various outdoor establishments are more exposed to such type of biological hazards.

38. How does dry chemical fire extinguisher work?

Ans. Dry chemical fire extinguishers extinguish the fire primarily by interrupting the chemical reaction of the fire triangle. Today's most widely used type of fire extinguisher is the multipurpose dry chemical that is effective on Class A, B and C fires. This agent also works by creating a barrier between the oxygen element and the fuel element on Class A fires. Ordinary dry chemical is for Class B and C fires only. It is important to use the correct extinguisher for the type of fuel. Using the incorrect agent can allow the fire to re-ignite after apparently being extinguished successfully.

LONG ANSWER TYPE QUESTIONS

39. Why is Internet called 'Network of Networks'?

Ans. Internet is called 'Network of Networks' because it is global network of computers that are linked together by cables and telephone lines making communication possible among them. It can be defined as a global network over a million of smaller heterogeneous computer networks.

The network which consists of thousands of networks spanning the entire globe is known as Internet. The Internet is a world wide collection of networked computers, which are able to exchange information with each other very quickly. Mostly people use the Internet in two ways, E-mail and World Wide Web. In Internet, most computers are not connected directly, they are connected to smaller networks, which in turn are connected through gateways to the Internet backbone. A gateway is a device that connects dissimilar networks. A backbone is central interconnecting structure - that connects one or more networks.

40. Give an example of E-mail address and explain each part of it.

Ans. The example of an E-mail address is abc@gmail.com. The format of E-mail address is username@hostname or domain name. So, as per the above example of E-mail address. abc is the username and gmail.com is the name of hosting server or host (domain) name. Thus, we can say that, E-mail address has two parts separated by '@' symbol.

- (i) Username On the left side of separator @ is the user name. A user name cannot have blanks.
- (ii) Domain Name for the Host Server The portion to the right of @ identifies the server or host network that services your E-mail. It is also known as E-mail server.

41. What is switching? Explain the different types of switching techniques.

Ans. Switching is a technique which is used in large networks. It is a hardware or software device which creates a connection between one or more computers. Switching is basically of three types which are as follows

- (i) Circuit Switching In this type of technique, an end-to-end path is created before sending the data (dedicated channel, i.e. Circuit). The data is transferred using physical connection which is set upto between sender and receiver.
- (ii) Message Switching Message switching is based on the transfer of block of messages from sender to receiver. Firstly, the address of destination is attached in a message. When a sender has a block of data to be sent, it is stored in the first switching station and then forwarded later. Each block of message is received entirely, checked for errors and then retransmitted. There is no need for physical connection between packet which has header that contains source and destination addresses information acknowledgement and error bits.
- (iii) Packet Switching In packet-based networks, the message gets broken into small data packets. These packets are sent out from the computer and they travel around the network seeking out the most efficient route to travel as circuits become available.

42. What are the preventive measures to avoid cyberbullying?

Ans. Preventive measures to be taken to avoid cyberbullying are given as

- (i) Check your privacy settings in social media like Facebook, so that your friends can only view your profile and posts.
- (ii) You should think twice while posting any message, as internet is public.

(iii) Always keep your personal information like name, address, phone number, school name, credit card or debit card credentials, password, etc., private.

43. What are some examples of workplace hazards?

Ans. Some examples of workplace hazards are as follows:

- (i) Damaged electrical outlets, cords, cables, etc.
- (ii) Overloaded outlets and circuits.
- (iii) Combustible substances in unsecured locations, (including excessive trash and recycling) - keep these far from electrical equipment.
- (iv) Fire exit obstacles. Keep workspace and equipment clean, dry and well-ventilated and especially clean of oil and dust.

44. What is a workplace emergency?

Ans. A workplace emergency is an unforeseen situation that threatens your employees, customers or the public; disrupts or shuts down the workplace operations; or causes physical or environmental damage.

Emergencies may be natural or manmade and include the following

- Floods
- Hurricanes
- Fires
- Tornadoes
- Toxic gas releases
- Chemical spills
- Radiological accidents
- Explosions
- Civil disturbances
- Workplace violence resulting in bodily harm and trauma

45. Write any five precautions that are to be taken at the workplace.

Ans. Any five precautions that are to be taken at the workplace are as follows:

- (i) Take responsibility and clean up, if you made a mess.
- (ii) Clean and organise your workspace.
- (iii) Ensure a clear and easy route to emergency exits and equipment.
- (iv) Be alert and awake on the job.
- (v) Be attentive at all times to your work surroundings.

46. Explain some of the ways to remove workplace hazards.

Ans. Some of the ways to eliminate workplace hazards are as follows

- (i) Remove damaged electrical outlets, cords, cables, etc.
- (ii) Check overloaded outlets and circuits.
- (iii) Combustible substances in unsecured locations, (including excessive trash and recycling) - keep these far from electrical equipment.
- (iv) Clear all fire exit obstacles.
- (v) Keep workspace and equipment clean, dry and well-ventilated and especially clean of oil and dust.
- (vi) Follow workspace protocol and guidelines to ensure safety and health; know and understand rules and procedures concerning fire emergencies.
- (vii) Ensure that smoke alarms and sprinkler systems are installed, working properly and are not blocked.
- (viii) Conduct regular fire drills.

47. Explain some of the actions that are used in the event of fire.

Ans. Some actions that should be used in the event of a fire are as follows:

- (i) Close all the windows, doors and switch of non-essential services.
- (ii) Walk to the nearest assembly point.
- (iii) Follow the emergency signs.
- (iv) Escape routes should be lit by emergency lighting so that people can use them safely if an emergency occurs.
- (v) Don't re-enter the building until permission is given by the person in charge of the evacuation.
- (vi) If the fire is small, but only if you are trained to do so use fire extinguishers to attempt to put the fire out.

48. Explain any five precautions at workplace.

Ans. Any five precautions at workplace are as follows:

- (i) First aid boxes at various places in the office.
- (ii) Trained first aiders - on all floors of a building.
- (iii) Knowledge of the nearest casualty department or local hospital.
- (iv) Access to call for an ambulance.
- (v) Ability to provide immediate assistance to casualties with illnesses or injuries caused by a specific hazard at work.

